Neogobius melanostomus (Round Goby)
Fishes
Exotic

Collection Info
Point Map
Species Profile
Animated Map
Impacts

Image 1 of 3



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Neogobius melanostomus (Pallas, 1814)

Common name: Round Goby

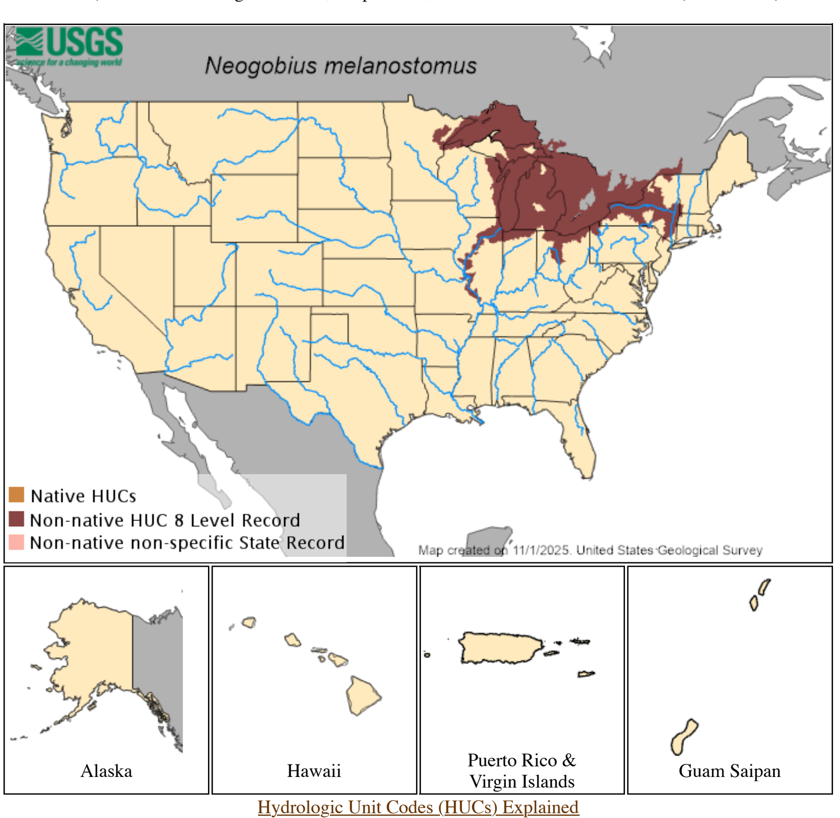
Synonyms and Other Names: *Apollonia melanostoma* (Pallas, 1814), *Apollonia melanostomus* (Pallas, 1814); see Stepien and Tumeo (2006) for name change, and Stepien and Neilson (2013) for clarification on taxonomy and nomenclature.

Taxonomy: available through ITIS

Identification: Distinguishing characteristics have been given by Berg (1949), Miller (1986), Crossman et al. (1992), Marsden and Jude (1995), and Hubbs et al. (2004). Young round gobies are solid slate gray. Older fish are blotched with black and brown and have a white to greenish dorsal fin with a black spot at the posterior base (but note some Lake Erie specimens are spotless). This goby is superficially similar to native sculpins but can be distinguished by the presence of fused pelvic fins (sculpins have two separate fins) (Marsden and Jude 1995; Hubbs et al. 2004).

Size: 30.5 cm; 17.8 cm maximum seen in United States

Native Range: Freshwater, prefers brackish (Stepien and Tumeo 2006). Eurasia including Black Sea, Caspian Sea, and Sea of Azov and tributaries (Miller 1986).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Neogobius melanostomus* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
IL	1993	2025	13	Cahokia-Joachim; Chicago; Des Plaines; Kankakee; Lake Michigan; Little Calumet-Galien; Lower Fox; Lower Illinois; Lower Illinois-Lake Chautauqua; Lower Illinois-Senachwine Lake; Peruque-Piasa; Pike-Root; Upper Illinois
IN	1993	2023	3	Chicago; Lake Michigan; Little Calumet-Galien
MI	1990	2025	46	Au Gres-Rifle; Au Sable; Betsie-Platte; Betsy-Chocolay; Birch-Willow; Black-Macatawa; Boardman-Charlevoix; Carp-Pine; Cass; Cedar-Ford; Cheboygan; Clinton; Detroit; Fishdam-Sturgeon; Flint; Huron; Kalamazoo; Kawkawlin-Pine; Lake Erie; Lake Huron; Lake Michigan; Lake St. Clair; Lake Superior; Little Calumet-Galien; Lone Lake-Ocqueoc; Lower Grand; Manistee; Manistique River; Menominee; Millecoquins Lake-Brevoort River; Muskegon; Ottawa-Stony; Pere Marquette-White; Pigeon-Wiscoggin; Pine; Raisin; Saginaw; Shiawassee; St. Clair; St. Joseph; St. Marys; Tacoosh-Whitefish; Thornapple; Thunder Bay; Tittabawassee; Upper Grand
MN	1996	2025	5	Baptism-Brule; Beartrap-Nemadji; Beaver-Lester; Lake Superior; St. Louis
MO	2023	2023	1	Upper Mississippi-Cape Girardeau
NY	1995	2025	27	Black; Buffalo-Eighteenmile; Cattaraugus; Chaumont-Perch; Chautauqua-Conneaut; Headwaters St. Lawrence River; Hudson-Hoosic; Hudson-Wappinger; Irondequoit-Ninemile; Lake Erie; Lake Ontario; Lower Genesee; Middle Hudson; Mohawk; Niagara River; Oak Orchard-Twelvemile; Oneida; Oswegatchie; Oswego; Owego-Wappasening; Raisin River-St. Lawrence River; Rondout; Salmon; Salmon-Sandy; Schoharie; Seneca; Upper Susquehanna
OH	1993	2025	12	Ashtabula-Chagrin; Black-Rocky; Cedar-Portage; Chautauqua-Conneaut; Cuyahoga; Grand; Huron-Vermilion; Lake Erie; Lower Maumee; Ottawa-Stony; Sandusky; Upper Scioto
PA	1996	2025	3	Chautauqua-Conneaut; French; Lake Erie
WI	1995	2025	13	Beartrap-Nemadji; Door-Kewaunee; Duck-Pensaukee; Lake Michigan; Lake Superior; Lower Fox; Manitowoc-Sheboygan; Menominee; Milwaukee; Oconto; Peshtigo; Pike-Root; St. Louis

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: The Round goby perches on rocks and other substrates in shallow areas, yet it has also been reported to flourish in a variety of habitat types including open sandy areas and in abundant aquatic macrophytes (Jude and DeBoe 1996; Clapp et al. 2001). Highest densities are reported between 1 and 50m depths, but the species has been reported out to 180m and is migrating offshore seasonally. Most reports are in coastal margins with high thermal energy, low exposure and high tributary influence. Round goby has a broad temperature tolerance but energetic optimum temperature of about 26° C and appears to avoid colder streams (Kornis and Vander Zanden 2010). This species has broad salinity tolerance surviving brackish conditions to 20ppt. It can tolerate low oxygen concentrations and poor water quality. Streams with low concentrations of dissolved ions such as calcium appear to resist invasion (Kornis et al 2013). Coulter et al. (2015) examined the influence of various habitat characteristics (e.g., wetlands vs open water, productivity, zooplankton and fish community diversity) on the abundance of Round goby at several sites in Lakes Michigan and Huron, finding that catch per unit effort was generally related to biological productivity but the direction and strength of the relationship varied across sites.

Female round gobies mature at 1-2 years with clutch (100-4000 eggs) related to body size. They can spawn every 20 days from April to September and have a maximum lifespan of about 4 years. Round goby has been shown to exhibit phenotypic plasticity in some life history characteristics, such as reproductive traits like number and size of mature oocytes, and this plasticity has been suggested to enhance establishment success at invasion fronts (Hôrková and Kovác 2015). Phenotypic plasticity has also been shown in Round goby to maintain or regain homeostasis after rapid temperature changes, thus increasing its invasion success (Wellband and Heath 2017).

The Round goby has a well-developed sensory system that enhances its ability to detect water movement. This allows it to feed in complete darkness, giving it an advantage over other fish in the same habitat (Wisconsin Sea Grant 2008). Round goby also forage efficiently in structured habitats that might otherwise serve as prey refugia (Gebauer et al 2019).

Round goby are dietary generalists feeding on both invertebrates – Dreissenid mussels, fingernail clams, small native mussels and snails, soft-shelled crayfish, Ephemeroptera, Plecoptera, Trichoptera, Chironomidae, Ceratapogonidae, Diptera, Odonata, Oligochaeta, Ostracoda, Isopoda, Amphipoda, Mysidae, and large Cladocerans (Jude et al 1992, Jude et al 1995, Ray and Corkum 1997, French and Jude 2001, Ricciardi 2001, Phillips et al 2003, Barton et al 2005, Hogan et al 2007, Walsh et al 2007, Krakowiak and Pennuto 2008, Polacik et al 2009, Kipp et al 2012) – as well as on small fishes and fish eggs (Crossman et al 1992, Jude et al 1995, Marsden and Jude 1995, Ray and Corkum 1997, Chotkowski and Marsden 1999, French and Jude 2001, Jude 2001, Ricciardi 2001, Hensler and Jude 2007, Gebauer et al 2018, Gebauer et al 2019, LeBlanc et al 2019). The zebra mussel (*Dreissena polymorpha*) may have facilitated the invasion of the Round goby by providing an abundant food source (Ricciardi and MacIsaac 2000).

Round gobies, particularly young gobies, are eaten by many native Great Lakes fishes including Lake Trout (*Salvelinus namaycush*), Smallmouth Bass (*Micropterus dolomieu*), Mottled Sculpin (*Cottus bairdii*), Northern Madtom (*Noturus stigmosus*), Burbot (*Lota lota*) and Lake Sturgeon (*Acipenser fulvescens*) (French and Jude 2001, Steinhart et al 2004, Dietrich et al 2006, Brey 2006, Hogan et al 2007, Taraborelli et al 2010, Madenjian et al 2011, Crane and Einhouse 2016, Jacobs et al 2017).

Round goby is one of the most frequently found prey items in the diet of the Double-Crested Cormorant (*Phalacrocorax auritus*) (Van Guilder and Seefelt 2013, Madura and Jones 2016). Round Gobies now comprise the majority of the diet for Lake Erie water snakes (*Nerodia sipedon insularum*), and the abundance of gobies has been credited for the increase in population size, increased growth rates, and larger body size of the snakes (King et al. 2006).

Means of Introduction: Introduced into the Great Lakes from the Black Sea via freighter ballast. Spread to Lake Superior by freighters operating within the Great Lakes (Hensler and Jude 2007); larval round goby have a diel vertical migration that facilitates them being taken in with ship ballast water and spread port to port. Round goby distribution in the Baltic Sea was partially determined by shipping activity, thus it is likely the same applies to the Great Lakes populations (Kotta et al. 2016). Evidence suggests that gobies make use of boat hulls as habitat, and could nest on boats and therefore be vectored to new areas via boat hulls (Bussmann and Burkhardt-Holm 2020).

Status: Widespread, overwintering and reproducing in all of the Great Lakes. Densities in some locations exceed 20 per square meter (Marsden and Jude 1995). Populations are expanding into tributaries (Phillips et al 2003, Krakowiak and Pennuto 2008, Kornis et al 2013, Bradshaw-Wilson 2019).

Established outside of the Great Lakes basin in 1994 (Dennison, personal communication), and in 2010 spread into the lower Illinois River (K. Irons, Illinois Natural History Survey, Champaign, IL, personal communication).

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological	Economic	Human Health	Other

The distribution of the Round Goby around the inshore areas of the Black and Caspian seas indicates their potential for widespread occupation of inshore habitats with cover, especially plants or rocky rubble, in the lower Great Lakes, yet they can migrate to deeper water 50-60 m in winter (Jude et al., 1992).

The numbers of native fish species have declined in areas where the Round Goby has become abundant (Crossman et al., 1992). This species has been found to prey on darters, other small fish, and Lake Trout eggs and fry in laboratory experiments. They also may feed on eggs and fry of sculpins, darters, and Logperch (Marsden and Jude, 1995) and have also been found to have a significant overlap in diet preference with many native fish species. French and Jude (2001) suggested that Round Goby competes with Rainbow Darter (*Etheostoma caeruleum*), Logperch (*Percina caprodes*), and Northern Madtom (*Noturus stigmosus*) for small macroinvertebrates; however, Burkett and Jude (2015) found that gobies exhibited little diet overlap with most small benthic fishes.

Mottled Sculpins (*Cottus bairdi*) have been particularly affected since the establishment of *N. melanostomus* (Marsden and Jude 1995). This is almost certainly due to competition with sculpins for spawning sites in large Round Goby (greater than 100 mm), for space in medium Round Goby (60-100 mm) and for food in small Round Goby (less than 60 mm) (Janssen and Jude 2001). Janssen and Jude (2001) argued that the main cause of the dramatic decline in the native Mottled Sculpin population is due to nesting interference with Round Goby; the other competition factors having a less severe impact, although they acknowledge the need for further research on food competition. Adults aggressively defend spawning sites and occupy prime spawning areas, keeping natives out (Marsden and Jude 1995; Dubs and Corkum 1996). Laboratory experiments have shown that the more aggressive *N. melanostomus* will evict *C. bairdi* from rock shelters that are being used for spawning or daytime predator evasion (Dubs and Corkum 1996). In trials where gobies were introduced into tanks with Mottled Sculpin residents, the gobies approached and chased the resident sculpin (Dubs and Corkum 1996). When sculpin were released into resident Round Goby tanks, the sculpin were chased and bitten (Dubs and Corkum 1996). Sculpin did not exhibit any aggressive behavior towards the gobies in any scenario (Dubs and Corkum 1996). In Calumet Harbor, there has been an absence of Mottled Sculpin nests and fish aged 0 since 1994, coinciding with *N. melanostomus* establishment (Janssen and Jude 2001). *Neogobius melanostomus* and *C. bairdi* both take daytime refuge from predators under rocks, emerging to feed nocturnally (Dubs and Corkum 1996). This space competition could displace *C. bairdi* into deeper and unprotected spaces where they can easily be predated. Competition for food between *N. melanostomus* and *C. bairdi* occurs most heavily when they are young (less than 60 mm), due to the overlap of an arthropod diet at this age (Janessen and Jude 2001).

The diet of larger Round Gobies consists mainly of zebra mussels, which no other fish species of the Great Lakes consumes so heavily, allowing round gobies to uniquely exploit a resource that could fuel a population explosion (Vanderploeg et al. 2002). In LeBoeuf and French creeks, Pennsylvania, native juvenile Unionid mussels comprised a significant portion of Round Goby diets (Bradshaw-Wilson et al. 2019). Walleye anglers in Detroit report that at times, all they can catch are gobies, which eagerly attack bait (Marsden and Jude 1995).

The invasion of Round Goby into Lake Erie has had very real environmental and economic impacts. Beginning in 2004, The State of Ohio has closed the Smallmouth Bass fishery in Lake Erie during the months of May and June, due to high predation rates by gobies on nests affecting recruitment. Under normal circumstances male Smallmouth Bass guard nests and are effective in deterring gobies. When males are removed, gobies immediately invade and have been shown to eat up to 4,000 eggs within 15 minutes. The months of May and June normally account for 50 percent of the total Smallmouth Bass catch in Lake Erie so there will be a considerable loss in funds generated by recreational fishers (National Invasive Species Council 2004).

The Menasha Lock on the Fox River (Wisconsin) is closed to navigation to prevent upstream movement of *N. melanostomus* into Lake Winnebago. An electric barrier has been proposed to allow the lock to open but still prevent upstream fish movement (Prinsen, 2021).

Neogobius melanostomus introductions may also be a vector for the spread of avian botulism. The change in behavior of infected gobies make them preferred prey items to piscivorous birds (Yule et al. 2006). At Lake Erie, botulism infected birds had been feeding more on Round Goby compared to uninfected birds (Corkum et al. 2004). Increased abundance of Round Goby in the diet of double-crested cormorant (*Phalacrocorax auritus*) may reduce chick growth and reproductive success, due to a lower energy density compared to other native fishes (Ruetz et al. 2009), and thus could provide some control over cormorant populations (Van Guilder and Seefelt 2013). Round Goby may provide an entry point for polychlorinated biphenyls (PCBs) into trophic webs. Macksasitorn et al. (2015) found a weak correlation between sediment and goby tissue PCB concentrations, with smaller gobies having higher PCB concentration.

Not all impacts of the introduced Round Goby are negative. Round Gobies comprise the majority of the diet for Lake Erie water snakes (*Nerodia sipedon insularum*), and the abundance of gobies has been credited for the increase in population size, increased growth rates, and larger body size of the snakes (King et al. 2006). Due to their increase in abundance, the Lake Erie water snake was removed from the federal Endangered Species List in 2011. In addition, round gobies provide an abundant food source for several sportfishes including walleye (Taraborelli et al. 2010), yellow perch (Truemper and Lauer 2005), and largemouth/smallmouth bass (Steinhart et al. 2004; Taraborelli et al. 2010). Round goby became the primary food resource for smallmouth bass in New York waters of Lake Erie following invasion, with a marked decrease in crayfish consumption, and bass had a larger length-at-age in the post-invasion period (Crane and Einhouse 2016). Jacobs et al. (2017) used stable isotope analysis to examine Lake Sturgeon (*Acipenser fulvescens*) in Lake Ontario and lower Nigara River pre- and post-invasion of Round Goby, finding an increased trophic position with age, an increased rate of $\delta^{15}N$ enrichment, and a marked importance of Round Goby in Lake Sturgeon diets in the post-invasion period. Jacobs et al. (2017) suggest that high abundance of this novel food source has altered Lake Sturgeon feeding ecology and may have increased population growth and success.

Remarks: Although the species exhibits two pigmentation morphs and investigations were planned to determine whether more than one introduction of *Neogobius* occurred in the Great Lakes (T. Cavender, Ohio State University, Columbus, OH, personal communication), only *N. melanostomus* has been observed.

Adrian-Kalchhauser et al. (2017) suggest that there is no direct evidence for the direct attachment of egg clutches by Round goby to aquatic vessels, a commonly suggested introduction vector for Ponto-Caspian gobiids, and that this concept should be treated as an untested hypothesis.

Pettitt-Wade et al. (2015) examined trophic niche breadth, plasticity, and overlap between Round and Tubenose gobies in Lakes Superior and St. Clair using stable isotope analysis. They found a higher isotopic trophic position and generally higher isotopic nichc breadth and plasticity in Round Goby, with little overlap between size-matched Round and Tubenose gobies, and suggested that this increased isotopic niche breadth and plasticity has assisted in the establishement success of Round Goby in the Great Lakes (widely abundant and distributed vs. low abundance and localized distribution of tubenose goby).

Introduced populations of Round Goby in the Great Lakes show reduced diversity and numbers of parasites compared to populations from its native range as well as to native Great Lakes fishes, providing some support for the 'enemy release hypothesis' for invasion success (Kvach and Stepien 2008; Gendron et al. 2012). However, Gendron et al. (2012) found that gobies from Lake St. Clair (one of the earliest introduced populations) showed an increase in parasite diversity and density over time. Similar results are apparent in Lake Erie (Kvach and Stepien 2008) and Lake Michigan (Camp et al 1999). Additional studies of introduced and native populations in Europe (Kvach et al 2014, Ondrackova et al 2015, Kvach et al 2017, Ovcharenko et al 2017, Mineeva 2019) also indicate that that enemy release is a transient effect seen only in early stages of invasion.

Round Goby has been shown to exhibit phenotypic plasticity in some life history characteristics, such as reproductive traits like number and size of mature oocytes, and this plasticity has been suggested to enhance establishment success at invasion fronts (Hôrková and Kovác 2015). Phenotypic plasticity has also been shown in Round Goby to maintain or regain homeostasis after rapid temperature changes, thus increasing its invasion success (Wellband and Heath 2017).

Voucher specimens: Ohio (OSM, UF 98888); Michigan (UMMZ); Illinois (INHS). Indiana (INHS, UMMZ 224874). Voucher specimens from the Canadian side of the St. Clair River (UMMZ 217682, 218279; ROM 60675); Lake Ontario (Ontario Federation of Anglers and Hunters, Peterborough, Ontario).

References (click for full references)

Adrian-Kalchhauser, I., A. N'Guyen, P.E. Hirsch, and P. Burkhardt-Holm. 2017. The invasive round goby may attach its eggs to ships or boats – but there is no evidence. Aquatic Invasions 12(2):263-267. http://www.aquaticinvasions.net/2017/AI_2017_Adrian-Kalchhauser_etal.pdf.

Alexander, J. 2004. Goby population explodes in Lakes. Everything Michigan (November 13, 2004) http://www.mlive.com/printer/printer.ssf?/base/news-5/1100344516168470.xml

Almqvist, G., A.K. Strandmark, and M. Appelberg. 2010. Has the invasive round goby caused new links in Baltic food webs? Environ Biol Fish 89:79-93.

Anonymous. - Ontario Ministry of Natural Resources, Niagara Falls, Ontario, Canada.

Balshine, S., A. Verma, V. Chant, and T. Theysmeyer. 2005. Competitive interactions between round gobies and logperch. Journal of Great Lakes Research 31(1):68-77.

Baltz, D.M., and P.B. Moyle. 1993. Invasion resistance to introduced species by a native assemblage of California stream fishes. Ecological Applications 3(2):246-255.

Barton, D.R., R.A. Johnson, L. Campbell, J. Petruniak, and M. Patterson. 2005. Effects of round gobies (*Neogobius melanostomus*) on dreissenid mussels and other invertebrates in eastern Lake Erie, 2002-2004. Journal of Great Lakes Research 31(2):252-261.

Berg, L.S. 1948-1949. Freshwater fishes of the U.S.S.R. and adjacent countries, 4th edition. Three volumes. Translated from Russian, 1962-1965, for the Smithsonian Institution and the National Science Foundation, by Israel Program for Scientific Translations, Jerusalem, Israel. Volume 1:504 pp.; volume 2:496 pp.; volume 3:510 pp.

Bergstrom, M.A., L.M. Evrard, and A.F. Mensinger. 2008. Distribution, abundance, and range of the round goby, *Apollonia melanostoma*, in the Duluth-Superior Harbor and St. Louis River Estuary 1998-2004. Journal of Great Lakes Research 34:535-543.

Bergstrom, M.A. and A.F. Mensinger. 2009. Interspecific Resource Competition between the Invasive Round Goby and Three Native Species: Logperch, Slimy Sculpin, and Spoonhead Sculpin. Transactions of the American Fisheries Society 138:1009-1017.

Bowen, A. - US Fish and Wildlife Service, Alpena, MI.

Bradshaw-Wilson, C., J. Stauffer, J. Wisor, K. Clark, and S. Mueller. 2019. Documentation of freshwater mussels (Unionidae) in the diet of round gobies *(Neogobius melanostomus)* within the French Creek watershed, Pennsylvania. The American Midland Naturalist 181(2):259-270. https://doi.org/10.1674/0003-0031-181.2.259

Brey, M.K. 2006. Changes in the Diet of Lake Trout (*Salvelinus namaycush*) in Near-Shore Lake Michigan with the Invasion of the Round Goby (*Neogobius melanostomus*): 1995-2005. Unpublished M.S. thesis. Eastern Illinois University, Illinois.

Burkett, E.M., and D.J. Jude. 2015. Long-term impacts of invasive round goby *Neogobius melanostomus* on fish community diversity and diets in the St. Clair River, Michigan. Journal of Great Lakes Research 41(3):862-872. http://dx.doi.org/10.1016/j.jglr.2015.05.004

Bussmann, K., and P. Burkhardt-Holm. 2020. Round gobies in the third dimension-use of vertical walls as habitat enables vector contact in a bottom-dwelling invasive fish. Aquatic Invasions 15(4):683-699. https://doi.org/10.3391/ai.2020.15.4.09.

Camp, J.W., L.M. Blaney, and D.K. Barnes. 1999. Helminths of the round goby, *Neogobius melanostomus* (Perciformes: Gobiidae), from southern Lake Michigan, Indiana. Journal of the Helminthological Society of Washington 66(1):70-72.

Cartwright, A., R. Gebauer, T. Vanina, V. Stejskal, B. Drozd. 2019. Shelter competition between mature non-indigenous western tubenose goby (*Proterorhinus semilunaris*) and immature invasive round goby (*Neogobius melanostomus*) for plants and rocks. Biol. Invasions 21:2723-2734.

Chotkowski, M.A., and J.E. Marsden. 1999. Round goby and mottled sculpin predation on lake trout eggs and fry: field predictions from laboratory experiments. Journal of Great Lakes Research 25(1):26-35.

Clapp, D.F., P.J. Schneeberger, D.J. Jude, G. Madison, and C. Pistis. 2001. Monitoring round goby (*Neogobius melanostomus*) population expansion in eastern and northern Lake Michigan. Journal of Great Lakes Research 27(3):335-341.

Clearwater, S.J., C.W. Hickey, and M.L. Martin. 2008. Overview of potential piscicides and molluscicides for controlling aquatic pest species in New Zealand. Science & Technical Publishing, New Zealand Department of Conservation, Wellington, New Zealand.

Corkum, L.D., M.R. Sapota, and K.E. Skora. 2004. The round goby, *Neogobius melanostomus*, a fish invader on both sides of the Atlantic Ocean. Biological Invasions 6:173-181.

Coulter, D.P., B.A. Murry, and D.G. Uzarski. 2015. Relationships between habitat characteristics and round goby abundance in Lake Michigan and Huron. Journal of Great Lakes Research 41(3):890-897. http://dx.doi.org/10.1016/j.jglr.2015.06.001

Crane, D.P., and D.W. Einhouse. 2016. Changes in growth and diet of smallmouth bass following invasion of Lake Erie by the round goby. Journal of Great Lakes Research 42(2):405-412. http://dx.doi.org/10.1016/j.jglr.2015.12.005

Crossman, E.J., E. Holm, R. Cholmondeley, and K. Tuininga. 1992. First record for Canada of the rudd, *Scardinius erythrophthalmus*, and notes on the introduced round goby, *Neogobius melanostomus*. Canadian Field-Naturalist 106(2):206-209.

Czypinski, G.D., A.K. Hintz, M.T. Weimer, and A. Dextrase. 2000. Surveillance for ruffe in the Great Lakes, 1999. U.S. Fish and Wildlife Service, Ashland, WI. 29 pp.

Czypinski, G.D., A.K. Bowen, M.T. Weimer, and A. Dextrase. 2002. Surveillance for ruffe in the Great Lakes, 2001. U.S. Fish and Wildlife Service, Ashland, WI. 36 pp.

Dietrich, J.P., B.J. Morrison, and J.A. Hoyle. 2006. Alternative ecological pathways in the eastern Lake Ontario food web—round goby in the diet of lake trout. Journal of Great Lakes Research 32(2):395-400.

Dubs, D.O.L., and L.D. Corkum. 1996. Behavioral interactions between round gobies (*Neogobius melanostomus*) and mottled sculpins (*Cottus bairdi*). Journal of Great Lakes Research 22:838-845.

Dunning, D.J., Q.E. Ross, E.T. Euston, and S.A. Haney. 2006. Association between the catches of round gobies and smallmouth bass on the Upper Niagara River. Journal of Great Lakes Research 32(4):672-679.

French, J.R.P, III and D.J. Jude. 2001. Diets and diet overlap of nonindigenous gobies and small benthic native fishes co-inhabiting the St. Clair River, Michigan. Journal of Great Lakes Research 27(3):300-311.

Gebauer, R., L. Veselý, A. Kouba, M. Buric, and B. Drozd. 2018. Forcasting impact of existing and emerging invasive gobiids under temperature change using comparative functional responses. Aquatic Invasions 13(2):289-297.

Gebauer, R. L. Vesely, T. Vanina, M. Buric, A. Kouba, and B. Drozd. 2019. Prediction of ecological impact of two alien gobiids in habitat structures of differing complexity. Can. J. Fish. Aquat. Sci 76:1954-1961.

Gendron, A.D., D.J. Marcogliese, and M. Thomas. 2012. Invasive species are less parasitized than native competitors, but for how long? The case of the round goby in the Great Lakes-St. Lawrence Basin. Biological Invasions 14:367-384.

GLMRIS. 2012. Appendix C: Inventory of Available Controls for Aquatic Nuisance Species of Concern, Chicago Area Waterway System. U.S. Army Corps of Engineers.

Hensler, S.R., and D.J. Jude. 2007. Diel Vertical Migration of Round Goby Larvae in the Great Lakes. Journal of Great Lakes Research 33(2):295-302.

Hintz, A. 2000. Lake Huron exotic fish surveillance in 1999. U.S. Fish & Wildlife Service, Fishery Resources Office, Alpena, Michigan. 13 pp.

Hogan, L.S., E. Marschall, C. Folt, and R.A. Stein. 2007. How non-native species in Lake Erie influence trophic transfer of mercury and lead to top predators. Journal of Great Lakes Research 33(1):46-61.

Hôrková, K., and V. Kovác. 2015. Ontogenetic phenomena, temporal aspect, and ecological factors in the successful invasion of round goby *Neogobius melanostomus* in the River Danube. Aquatic Invasions 10(2):227-235. http://dx.doi.org/10.3391/ai.2015.10.2.11

Hubbs, C.L., K.F. Lagler, and G.R. Smith. 2004. Fishes of the Great Lakes Region. Revised edition. University of Michigan Press, Ann Arbor, MI.

International Joint Commission. 2011. 2009-2011 Priority Cycle Report on Binational Aquatic Invasive Species Rapid Response. Prepared by the Binational Aquatic Invasive Species Rapid Response Work Group for the International Joint Commission. Canada and the United States.

Jacobs, G.R., E.L. Bruestle, A. Hussey, D. Gorsky, and A.T. Fisk. 2017. Invasive species alter ontogenetic shifts in the trophic ecology of Lake Sturgeon (*Acipenser fulvescens*) in the Niagara River and Lake Ontario. Biological Invasions 19:1533-1546. https://link.springer.com/article/10.1007/s10530-017-1376-6

Janssen, J., and D.J. Jude. 2001. Recruitment failure of mottled sculpin *Cottus bairdi* in the Calumet Harbor, southern Lake Michigan, induced by the newly introduced round goby *Neogobius melanostomus*. Journal of Great Lakes Research 27(2):319-328.

Johnson, T.B., D.B. Bunnell, and C.T. Knight. 2005. A potential new energy pathway in Central Lake Erie: the round goby connection. Journal of Great Lakes Research 31(Suppl. 2):238-251.

Jude, D.J. - University of Michigan and Freshwater Physicians, Inc. Ann Arbor, MI.

Jude, D. J., 2001. Round and tubenose gobies: 10 years with the latest Great Lakes phantom menace. Dreissena 11, 1-14

Jude, D.J., and S.F. DeBoe. 1996. Possible impact of gobies and other introduced species on habitat restoration efforts. Canadian Journal of Fisheries and Aquatic Sciences 53(S1):136-141.

Jude, D.J., R.H. Reider, and G.R. Smith. 1992. Establishment of Gobiidae in the Great Lakes Basin. Canadian Journal of Fisheries and Aquatic Sciences 49:416-421.

Jude, D.J., J. Janssen, and G. Crawford. 1995. Ecology, distribution, and impact of the newly introduced round and tubenose gobies on the biota of the St. Clair and Detroit Rivers. Pages 447-460 in Munawar, M., T. Edsall, and J. Leach, eds. The Lake Huron Ecosystem: Ecology, Fisheries, and Management. SPB Academic Publishing. Amsterdam, The Netherlands.

Jude, D.J., S.R. Hensler, M.M. Murray. 2018. Round goby and zebra mussel interactions with darters in a warm-water stream community in southern Michigan, USA. Journal of Freshwater Ecology 33(1):395-412.

Keppner, S. - US Fish and Wildlife Service, Amherst, NY.

King, R.B., J.M. Ray, and K.M. Stanford. 2006. Gorging on gobies: beneficial effects of alien prey on a threatened vertebrate. Canadian Journal of Zoology 84:108-115.

Kipp, R., I. Hebert, M. Lacharite, and A Ricciardi. 2012. Impacts of predation by the Eurasian round goby (*Neogobius melanostomus*) on molluscs in the upper St. Lawrence River. Journal of Great Lakes Research 38(1):78-89.

Knight, C. 1994. The round goby: Lake Erie's newest invader. Ohio Chapter American Fisheries Society (OCAFS) Newsletter 21(3):5.

Kornis, M.S., and M.J. Vander Zanden. 2010. Forecasting the distribution of the invasive round goby (*Neogobius melanostomus*) in Wisconsin tributaries to Lake Michigan. Can. J. Fish. Aquat. Sci 67:553-562.

Kornis, M.S., N. Mercado-Silva, and M.J. Vander Zanden. 2012. Twenty years of invasion: a review of round goby *Neogobius melanostomus* biology, spread and ecological implications. Journal of Fish Biology 80:235-285.

Kornis, M.S., S. Sharma, and M.J. Vander Zanden. 2013. Invasion success and impact of an invasive fish, round goby, in Great Lakes tributaries. Diversity and Distributions 19:184-198.

Kotta, J., K. Nurkse, R. Puntila, and H. Ojaveer. 2016. Shipping and natural environmental conditions determine the distribution of the invasive non-indigenous round goby *Neogobius melanostomus* in a regional sea. Estuarine, Coastal and Shelf Science 169:15-24.

Krakowiak, P.J., and C.M. Pennuto. 2008. Fish and macroinvertebrate communities in tributary streams of Eastern Lake Erie with and without round gobies (*Neogobius melanostomus*, Pallas 1814). Journal of Great Lakes Research 34(4):675-689.

Kuhns, L.A., and M.B. Berg. 1999. Benthic invertebrate community responses to round goby (*Neogobius melanostomus*) and zebra mussel (*Dreissena polymorpha*) invasion in southern Lake Michigan. Journal of Great Lakes Research 25(4):910-917.

Kvach, Y., and C.A. Stepien. 2008. Metazoan parasites of introduced round and tubenose gobies in the Great Lakes: support for the "enemy release hypothesis". Journal of Great Lakes Research 34:23-35.

Kvach, Y., Y. Kornyychuk, K. Mierzejewska, N. Rubtsova, V. Yurakhno, J. Grabowska, M. Ovcharenko. 2014. Parasitization of invasive gobiids in the eastern part of the Central trans-European corridor of invasion of Ponto-Caspian hydrobionts. Parisitol Res preprint:20.

Kvach, Y., M. Ondrackova, M. Janac and P. Jurajda. 2017. The parasite community of round goby *Neogobius melanostomus* (Pallas, 1814) (Actinopterygii: Gobiidae) newly introduced into the upper Elbe. Knowledge and Management of Aquatic Ecosystems 418(19):1-6.

Lauer, T.E., P.J. Allen, and T.S. McComish. 2004. Changes in mottled sculpin and johnny darter trawl catches after the appearance of round gobies in the Indiana waters of Lake Michigan. Transactions of the American Fisheries Society 133(1):185-189.

Leblanc, J.P., C.C. Killourhy, and J.M. Farrell. 2019. Round goby (*Neogobius melanostomus*) and native fishes as potential nest predators of centrarchid species in the upper St. Lawrence River. Journal of Great Lakes Research 1. https://doi.org/10.1016/j.jglr.2019.12.001.

Lederer, A., J. Massart, and J. Janssen. 2006. Impact of round gobies (*Neogobius melanostomus*) on dreissenids (*Dreissena polymorpha* and *Dreissena bugensis*) and the associated macroinvertebrate community across and invasion front. Journal of Great Lakes Research 32(1):1-10.

MacInnis, A.J., and L.D. Corkum. 2000. Age and growth of round goby *Neogobius melanostomus* in the upper Detroit River. Transactions of the American Fisheries Society 129(3):852-858.

MacInnis, A.J., and L.D. Corkum. 2000. Fecundity and reproductive season of the round goby *Neogobius melanostomus* in the upper Detroit River. Transactions of the American Fisheries Society 129(1):136-144.

Macksasitorn, S., J. Janssen, and K.A. Gray. 2015. PCBs refocused: Correlation of PCB conentrations in Green Bay legacy sediments with adjacent lithophilic, invasive biota. Journal of Great Lakes Research 41:215-221. http://dx.doi.org/10.1016/j.jglr.2014.12.021

Madenjian, C.P., M.A. Stapanian, L.D. Witzel, D.W. Einhouse, S.A. Pothoven, and H.L. Whitford. 2011. Evidence for predatory control of the invasive round goby. Biological Invasions 13(4):987-1002. http://link.springer.com/article/10.1007/s10530-010-9884-7

Madura, P.T., and H.P. Jones. 2016. Invasive species sustain double-crested cormorants in southern Lake Michigan. Journal of Great Lakes Research 42(2):413-420.

Marsden, J.E., and D.J. Jude. 1995. Round gobies invade North America. Fact sheet produced by Sea Grant at Ohio State University, Columbus, OH.

Miller, P.J. 1986. Gobiidae. Pages 1019-1085 in P. J. P. Whitehead, M.L. Bauchot., J.C. Hureau, J. Nielsen, E. Tortonese, editors. Fishes of the north-eastern Atlantic and the Mediterranean, volume III. United Nations Educational, Scientific and Cultural Organization, Paris, France.

Mineeva, O.V. 2019. The Trematode Fauna of Ponto-Caspian Gobies (Pisces, Gobiidae) in the Saratov Reservoir. Russian Journal of Biological Invasions 10(1):22-29.

Morrison, H.A., D.M. Whittle, and G.D. Haffner. 2000. The relative importance of species invasions and sediment disturbance in regulating chemical dynamics in western Lake Erie. Ecological Modelling 125(2-3):279-294.

Nagelkerke, L.A., E. van Onselen, N. van Kessel, and R.S.E.W. Leuven. 2018. Functional feeding traits as predictors of invasive success of alien freshwater fish species using a food-fish model. Plos-One Open Access:13. https://doi.org/10.1371/journal.pone.0197636.

National Invasive Species Council. 2004. Weekly Notice May 27, 2004-June 3, 2004.

Ng, C.A., M.B. Berg, D.J. Jude, J. Janssen, P.M. Charlebois, L.A.N. Amaral, and K.A. Gray. 2008. Chemical amplification in an invaded food web: Seasonality and ontogeny in a high-biomass, low-diversity ecosystem. Environmental Toxicology and Chemistry 27(10):2186-2195.

Ochs, C.L., A.J. Laframboise, W.W. Green, A. Basilious, T.B. Johnson, and B.S. Zielinski. 2013. Response to putative round goby (*Neogobius melanostomus*) pheromones by centrarchid and percid fish species in the Laurentian Great Lakes. Journal of Great Lakes Research 39:186-189.

Ondrackova, M., I. Hudcova, M. Davidova, Z. Adamek, M. Kasny, and P. Jurajda. 2015. Non-native gobies facilitate the transmission of *Bucephalus polymorphus* (Trematoda). Parasites and Vectors 8:382.

Ondrackova, M., Z. Valova, I. Hudcova, V. Michalkova, A. Simkova, J. Borcherding, and P. Jurajda. 2015. Temporal effects on host-parasite associations in four naturalized goby species living in sympatry. Hydrobiologia 746:233-243.

Ovcharenko, M. P. Wroblewski, Y. Kvach, O. Drobiniak. 2017. Study of *Loma acerinae* (Microsporidia) detected from three Ponto-Caspian gobies (Gobiidae) in Ukraine. Parisitology Research

Pettitt-Wade, H., K.W. Wellband, D.D. Heath, and A.T. Fisk. 2015. Niche plasticity in invasive fishes in the Great Lakes. Biological Invasions 17:2565-2280. http://link.springer.com/article/10.1007/s10530-015-0894-3

Phillips, E.C., M.E. Washek, A.W. Hertel, and B.M. Niebel. 2003. The round goby (*Neogobius melanostomus*) in Pennsylvania tributary streams of Lake Erie. Journal of Great Lakes Research 29(1):34-40.

Polacik, M., M. Janac, P. Jurajda, Z. Adamek, M. Ondrackova, T. Trickova, and M. Vassilev. 2009. Invasive gobies in the Danube: invasion success facilitated by availability and selection of superior food resources. Ecology of Freshwater Fish 18:640-649.

Poos, M., A.J. Dextrase, A.N. Schwalb, and J.D. Ackerman. 2010. Secondary invasion of the round goby into high diversity Great Lakes tributaries and species at risk hotspots: potential new concerns for endangered freshwater species. Biological Invasions 12:1269-1284.

Prinsen, J. 2021. Menasha lock remains closed due to invasive round goby. The Post-Crescent. Appleton, WI. May 18, 2021. https://www.postcrescent.com/story/news/2021/05/18/fox-river-locks-system-menasha-remains-closed-invasive-species/5145561001/. Created on 05/18/2021. Accessed on 05/19/2021.

Ray, W.J., and L.D. Corkum. 1997. Predation of zebra mussels by round gobies, *Neogobius melanostomus*. Environmental Biology of Fishes 50(3):267-273.

Ricciardi, A. 2001. Facilitative interactions among aquatic invaders: is an "invasional meltdown" occurring in the Great Lakes? Canadian Journal of Fisheries and Aquatic Sciences 58:2513-2525.

Ricciardi, A., and H.J. MacIsaac. 2000. Recent mass invasion of the North American Great Lakes by Ponto-Caspian Species. Trends in Ecology and Evolution 15:62-65.

Rollo, A., G. Andraso, J. Janssen, and D. Higgs. 2007. Attraction and localization of round goby (*Neogobius melanostomus*) to conspecific calls. Behaviour 144:1-21.

Ruetz, C.R., III, D.L. Strouse, and S.A. Pothoven. 2009. Energy density of introduced round goby compared with four native fishes in a Lake Michigan tributary. Transactions of the American Fisheries Society 138(4):938-947.

Savino, J.F., D.J. Jude, and M.J. Kostich. 2001. Use of electric barriers to deter movement of round goby. American Fisheries Society Symposium 26:171-182.

Steinhart, G.B., R.A. Stein, and E.A. Marschall. 2004. High growth rate of young-of-the-year smallmouth bass in Lake Erie: a result of the round goby invasion? Journal of Great Lakes Reserach 30:381-389.

Stepien, C.A., and M.E. Neilson. 2013. What's in a name? Taxonomy and nomenclature of invasive gobies in the Great Lakes and beyond. Journal of Great Lakes Research 39(4):555-559. http://dx.doi.org/10.1016/j.jglr.2013.09.020

Stepien, C.A., and M.A. Tumeo. 2006. Invasion genetics of Ponto-Caspian gobies in the Great Lakes: a 'cryptic' species, absence of founder effects, and comparative risk analysis. Biological Invasions 8:61-78.

Sun, X., T.B. Johnson and K.G. Drouillard. 2016. Determination of PCB Elimination Coefficients in Round Goby and Tubenose Goby. Bull. Environ. Contam. Toxicol 97:346-352.

Taraborelli, A.C., M.G. Fox, T.B. Johnson, and T. Schaner. 2010. Round goby (*Neogobius melanostomus*) population structure, biomass, prey consumption, and mortality from predation in the Bay of Quinte, Lake Ontario. Journal of Great Lakes Research 36:625-632.

Thompson, H.A., and T.P. Simon. 2014. Diet shift response in round goby, *Neogobius melanostomus*, based on size, sex, depth, and habitat in the western basin of Lake Erie. Journal of Applied Ichthyology 30(5):955-961.

Tilmant, J.T. 1999. Management of nonindigenous aquatic fish in the U.S. National Park System. National Park Service. 50 pp.

Truemper, H.A., and T.E. Lauer. 2005. Gape limitation and piscine prey size-selection by yellow perch in the extreme southern area of Lake Michigan, with emphasis on two exotic prey items. Journal of Fish Biology 66:135-149.

U.S. Environmental Protection Agency (USEPA). 2008. Predicting future introductions of nonindigenous species to the Great Lakes. EPA/600/R-08/066F. National Center for Environmental Assessment, Washington, DC, 138 pp. http://cfpub.epa.gov/ncea/cfm/recordisplay.cfm?deid=190305

U.S. Fish and Wildlife Service (USFWS). 2005. National Wildlife Refuge System Invasive Species. http://www.nwrinvasives.com/index.asp (Last accessed 2006)

Vanderploeg, H.A., T.F. Nalepa, D.J. Jude, E.L. Mills, K.T. Holeck, J.R. Leibig, I.A, Grigorovich, and H. Ojaveer. 2002. Dispersal and emerging ecological impacts of Ponto-Caspian species in the Laurentian Great Lakes. Canadian Journal of Fisheries and Aquatic Sciences 59:1209-1228.

Van Guilder, M.A., and N.E. Seefelt. 2013. Double-crested (*Phalacrocorax auritus*) chick bioenergetics following round goby (*Neogobius melanostomus*) invasion and implementation of cormorant population control. Journal of Great Lakes Research 39:153-161.

Weber, M.J., J.M. Dettmers, D.H. Wahl, and S.J. Czesny. 2011. Size preferences and behaviors of native yellow perch foraging on invasive round gobies. Journal of Great Lakes Research 37(3): 584-587.

Wellband, K.W., and D.D. Heath. 2017. Plasticity in gene transcription explains the differential performance of two invasive fish species. Evolutionary Applications(10):563-576. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5469171/pdf/EVA-10-563.pdf.

Wisconsin Sea Grant. 2008. Fish of the Great Lakes: round goby. http://www.seagrant.wisc.edu/greatlakesfish/roundgoby.html

Yule, A.M., I.K. Barker, J.W. Austin, and R.D. Moccia. 2006. Toxicity of *Clostridium botulinum* Type E neurotoxin to Great Lakes fish: implications for avian botulism. Journal of Wildlife Diseases 42(3):479-493.

Zuerwink, A., Jude, D., Gannon, J., 2019. Behavioral interactions between the nonindigenous round goby and young-of-the-year yellow perch. Bioinvasions 21, 3633-3639.

Other Resources:
Neogobius spp. [goby] (ANS Clearinghouse Bibliography)

Great Lakes Indian Fish and Wildlife Commission Maps

Great Lakes Waterlife

Global Invasive Species Database Factsheet

US Fish and Wildlife Service Ecological Risk Screening Summary for *Neogobius melanostomus*

FishBase Summary

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